Mohamad Bitar

AI Engineer

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SUMMARY

AI engineer who builds applied ML products: retrieval-augmented generation, text embeddings, vector search, and LLM systems that answer from real data with source citations. Took BAC AI from laptop experiments to a live website answering multilingual (Arabic, English, French) study questions from official textbooks. Fourth-year Informatics Engineering student specializing in AI, seeking AI/ML engineering internships and junior roles.

PROJECTS

BAC AI — Syrian Baccalaureate Study Assistant July 2026 – Present

Founder / AI Engineer | Python, FastAPI, Qdrant, BGE-M3, Qwen, React, Docker

- Built and launched a live study assistant where students chat with official Grade 12 textbooks, open the cited page, and revise with auto-generated quiz questions drawn from retrieved lessons.
- Designed the RAG pipeline end to end: PDF text extraction and OCR (Tesseract, EasyOCR), lesson-boundary detection, chunking strategy, BGE-M3 1,024-dim embeddings, and Qdrant vector search.
- Tuned retrieval as the main quality lever with intent routing before generation and query-dependent search (explain vs. solve, named page/lesson, Arabic grammar and poetry, English workbook vs. student book, French writing topics).
- Enforced grounding with separate scientific/literary catalogs, teacher guides for worked solutions, and citations pinned to the official textbook; evaluated answer quality by testing real textbook questions.
- Engineered prompts, conversation handling, and streaming generation on Qwen, served via Ollama locally and OpenRouter in production with compatible embeddings.
- Shipped the full product: FastAPI backend with JWT and Google sign-in, subscription access, admin panel, and a bilingual React app (Arabic RTL + English); migrated from an initial Flutter prototype to web; deployed with Docker and GitHub Actions on an Oracle Cloud VM, with API tests in pytest.

Heart Disease Prediction

Machine Learning Project | Python, scikit-learn, pandas

- Built a binary classification model predicting whether a patient has heart disease from several clinical factors.
- Trained and compared multiple models (logistic regression, random forest) to select the best performer.

SKILLS

AI Engineering: RAG, embeddings, vector search, prompt design, LLM integration, streaming generation, source citation, multilingual (Arabic/English/French) QA

Machine Learning: Text embeddings (BGE-M3), semantic search, chunking strategy, OCR, intent routing, binary classification, model comparison, evaluation

Models & Tools: Qwen, Ollama, OpenRouter, Qdrant, sentence-transformers, scikit-learn, pandas, Tesseract, EasyOCR

Languages: Python, TypeScript, SQL, Dart

Backend: FastAPI, Pydantic, SQLAlchemy, SQLite, JWT, OAuth (Google), REST APIs

Frontend: React, Vite, bilingual RTL UI

DevOps: Git, GitHub Actions, Docker, Linux

EDUCATION

Arab International University Expected 2028

Informatics Engineering, Artificial Intelligence specialization (fourth-year student)